[M] Multi-agent Simulation of Unpredictable Traffic Situations in a Simplified Model of Gothenburg

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An agent-based traffic simulation of the Gothenburg road network was developed to investigate how road closures affect urban traffic flow. The road network was modeled as a graph constructed from real-world data, and vehicles were represented as agents following a simple car-following model. The simulation was used to study the effect of varying traffic demand and the fraction of closed roads over a simulated day. The results show that traffic performance degrades as the number of closed roads increases, and this small-scale project demonstrates that agent-based simulation can be used to study traffic dynamics within a simplified urban traffic network.

Project Topic: The Fast and the Gridlocked

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I. INTRODUCTION

With constantly appearing construction sites, road redirections, and confusing traffic signs, Gothenburg is known for not being the most car-friendly city in Sweden. The unpredictable nature of the traffic in the city can often cause traffic jams, road rage and delays. With this as background, this project aims to model road traffic in a simplified road network based on Gothenburg and investigate how the closure of roads, for example due to construction, affects traffic flow. The central research question of the project is *"How do road closures affect congestion levels in a simplified model of the road network of Gothenburg?"*.

Traffic simulation is a broad subject and has become increasingly relevant over the past decades. With increased demand for travel and shipping, along with many different modes of transportation and complex road networks, our everyday traffic has become an incredibly complex system. And with the recent rapid developments in computing, simulation and deeper analysis of such systems has become possible.

A common method of simulating the emergent complex behavior of traffic is to use *agent-based* simulation models[1]. Such models rely on the use of many simple entities, called agents, and allowing them to interact in an environment. This structure will be the basis for the model implemented in this project to simulate traffic in Gothenburg.

II. OVERVIEW

Being such a large subject, it is useful to divide the subject of traffic simulation into different subfields: *macroscopic*, *microscopic*, and *nanoscopic* simulation[1, 2]. An overview of the three subfields is presented in table I. Which one, or mix, of these subfields you choose to study depends on which aspects of real world traffic dynamics you wish to model. If we consider, for example, transportation planning, you would want to simulate over a long timescale and large traffic networks, i.e., *macroscopic* simulation. On the other hand, if you were to model split-second decision making of an autonomous vehicle, you would work at a whole different scale of time and space and with other inputs to consider, requiring a completely different method, i.e., *nanoscopic* simulation.

Macroscopic traffic simulation. This method relies on high level mathematical models with dynamic relations between quantities such as traffic density, flow, and mean speed that vary across space and time[2, Ch. 6-9]. This description is essentially analogous to the equations describing liquids or gasses in motion, such as the continuity equation.

In this project, macroscopic quantities such as mean speed and mean congestion will be analyzed based on data generated by microscopic simulation.

Microscopic traffic simulation. This method introduces individual drivers, modeling their dynamics individually using more or less complex models. Here, agent based methods are used predominantly. One interesting approach is the use of so called *intelligent agents*, with behavioral patterns mimicking those of a human. In [3], the author created agents with beliefs, intentions, and preferences based on results from a behavioral survey they conducted. While this is outside the scope of this project, it is an interesting approach that inspired a small behavioral element later on in the project, see chapter III C.

Microscopic simulation also sometimes introduces a more complex road network with multiple lanes, complex intersections, and networks extend beyond 2D. Since the model in this project is being developed from scratch, and due to limitations in the data, the complexity of the road network model will be highly limited.

Nanoscopic traffic simulation. Nanoscopic simulation is a highly detailed simulation taking into account both internal dynamics, such as engine control and gear shifting, and reacting to external parameters such as road condition, lane markings, traffic signs, and other vehicles. It is an increasingly relevant topic with autonomous systems in vehicles becoming more common, but such detail in simulation will not be relevant for this project.

TABLE I: **Overview of different methods (or subfields) of traffic simulation.** The methods' different use cases, features and whether or not they are suitable for this project is listed briefly. For more in-depth discussion regarding the methods, see chapter II.

Method / Model	Use case scenario	Features	Suitable for the project?
Macroscopic	Large scale simulation, making predictions, logistics	Low level of detail (fewer parameters), computationally cheap	While the simulation will be more microscopic, macroscopic identities will be studied
Microscopic	Modeling of individual driver behavior, car-following models, analyzing urban traffic with multiple lanes and traffic signals	Higher level of detail, often agent based, cen be computationally expensive depending on detail	Some microscopic behavior will have to be incorporated to simulate the urban traffic situation of Gothenburg
Nanoscopic	Driver assistance systems, autonomous driving	Highly detailed simulation, incorporates internal vehicle functions, computer vision	This high detail of simulation will not be relevant for the project

III. METHOD

The method chosen for the project is an agent-based implementation of a mix between micro- and macroscopic traffic simulation. An agent can be seen as an abstract member of a simulation, often interacting with other agents. The method provides an intuitive way of modeling complex and dynamical behavior. In this case, our agents will be the cars traversing our road network. The road network is modeled using a directed graph data structure, where intersections function as nodes and roads as weighted edges. The use of this structure means that we can employ standard graph search algorithms for navigation, such as the A^*-algorithm (see III B)

A. Road network as a graph

To model a real-world road network as nodes and edges in a computer program, one needs real-world data. The data used in this project for modeling Gothenburg was provided by the Swedish Transport Administration (swe: Trafikverket) [4]. The data was packaged as a `GeoPackage` file (`.gpkg`), where road segments were listed along with coordinates. Using the Python library `GeoPandas` [5], one can easily extract the data and then use it to build the road network-graph. Road objects (i.e., edge objects) can then be given additional attributes such as length, speed limits and whether or not the road is open for traffic. For a complete graphical view of the road network, see figure 1.

Here comes a significant delimitation that had to be done in the project regarding the road network. Due to the data availability at the time, only coordinates and road lengths were extracted and used. This meant that other characteristics of the road network, such as number of lanes, road directionality, speed limit, and traffic light placement, had to be omitted. For simplicity, all roads were assumed to have one lane, be bi-directional, and have a speed limit of 50km/h.

B. A graph search algorithm

For vehicle routing (or pathfinding), graph search algorithms are an obvious choice. One of the most common algorithms for finding shortest paths is the A^*-algorithm. It is a version of Dijkstra's algorithm, utilizing a priority queue and cost-to-goal estimation to quickly find the shortest path from a start node to a destination. It is relatively computationally inexpensive as long as the graph is not too large, which was not the case in this project.

In this implementation, the cost of a path is the estimated time to travel along the path had there been no traffic. This equates to the sum of all road segments divided by the speed limit on that road. In this case, since the speed limit is constant everywhere, this will, however, have no practical difference from using path length as a cost function. The algorithm also assumes that all roads are open for travel, which was a conscious decision to introduce unpredictability in the simulation.

C. Modeling vehicles as agents

As mentioned, the vehicles in the simulation are modeled as agents traversing the road network. Each agent is simulated with their own physical variables (position, speed, acceleration) and abide to certain rules when interacting with their environment or other agents, and this is where the power of agent based simulation lies. Even though each agent is using a relatively simple deterministic model, complex global behavior occur when they interact. The goal of the project was to see such complex behavior in the context of traffic, such as traffic jams.

The main part of agent-to-agent interaction in this project is a so called "car-following model" [2, pp. 157-173]. It is a mathematical model of how a driver accelerates or decelerates depending on the distance to the vehicle in front. One such model is the *Full Velocity Difference Model* (FVDM), see [2, p. 171]. Given the gap s and the speed difference Δv between the two drivers, we

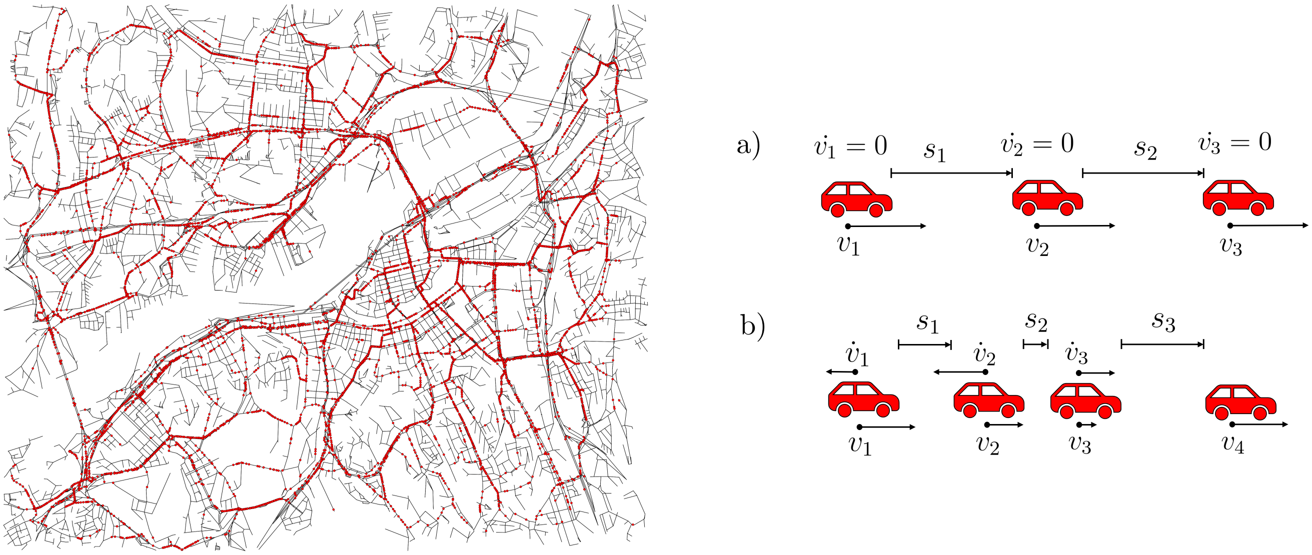

FIG. 1: **Microscopic traffic simulation of Gothenburg.** Left: Graphical representation of the road network of a portion of Gothenburg. Here vehicles are represented as red dots and roads as gray lines. Figure generated using Python package `pyqtgraph`. Right: Simplified figure of the car-following model (FVDM) at work and how vehicles react to each other. a) Traffic flows perfectly and all vehicles travel att constant speed. b) A congestion wave is starting to form and vehicles have to brake to avoid collision.

can model the acceleration as

$$\dot{v} = \frac{v_{\text{opt}} - v}{\tau} - \frac{\gamma \Delta v}{\max\left[1, s/(v_0 T)\right]}, \tag{1}$$

where τ, γ, v_0, and T are parameters related to the driver. τ [s] is called the *speed adaptation parameter* and measures how quickly the driver adapts to their desired speed. γ [s^{-1}] is the speed difference sensitivity, and v_0 [m/s] is the desired free flow speed, i.e., the speed the driver wants to drive when there is no one in front. Lastly, T is the desired time gap the driver wants to the leader. $v_{\text{opt}}(s)$ is the *optimal velocity function* and dictates how fast the driver wants to go given the gap s:

$$v_{\text{opt}}(s) = \max\left[0, \min\left(v_0, \frac{s - s_0}{T}\right)\right],$$

where s_0 is the minimum distance between vehicles at a standstill (bumper to bumper traffic). For a simple example figure of FVDM in action, see Fig. 1.

In the real world, different drivers behave differently in traffic. Some like to drive faster than the speed limit, some slower. Some choose to stay very close to the vehicle in front, others stay farther away. To introduce this variability in the drivers, a simple stochastic element was added, an "aggression" parameter α. α is assigned randomly in the range 0 to 1 when a vehicle is initialized and is then used to set the parameters in eq. 1, as well as the maximum acceleration and deceleration allowed. In table II, an overview of the parameters influenced by the aggression parameter and their respective ranges is presented. Here c_v is introduced, which is a "speed compliance" parameter that is used to calculate the desired speed $v_0 = c_v \cdot v_{limit}$.

TABLE II: **Ranges for driver behavior parameters used in the car following model.** Note that extremes correspond to very low or high aggression parameter α. Ranges were set based on both typical values from [2] and trial-and-error.

Parameter	Range	Unit
τ	1.5 – 4	s
c_v	0.85 – 1.15	–
γ	0.4 – 0.8	s^{-1}
T	1 – 3	s
a_{max}	1 – 4	m/s^2
b_{max}	3 – 7	m/s^2

D. The simulation

With the road network-graph, A^*-algorithm, and the driver logic implemented, the only thing remaining is to put it all together in a simulation. During initialization, the road network graph is created. To simulate construction sites, or otherwise closed roads, roads are set as closed randomly with a given percentage, p. After the graph is initialized, a set number of vehicles (agents) are spawned randomly on the network and simulation begins.

When a vehicle is spawned, it receives a random destination and uses the A^* algorithm to find the fastest path towards it, producing a planned route. The vehicle then tries to follow this route, updating their velocity and position every time step Δt according to FVDM (see eq. 1). If the vehicle encounters a closed road, it will try to find a new route to its destination. Eventually, the vehicle reaches its destination and is despawned.

This process is performed with every vehicle on the network at every time step Δt for a total time of T_{tot}. Each Δt there is also a probability of spawning new vehicles. This probability depends on the time of day and is based on data from a traffic survey conducted in Gothenburg on peoples travel habits[6].

During the simulation, variables such as the number of vehicles on the network, average velocity of all vehicles, and number of vehicles at a standstill are tracked. Likewise, when a vehicle completes their route, data such as estimated time to destination, actual time to destination, and number of reroutes are logged.

All simulation parameters not listed here and implementation details required for full reproducibility are provided in the accompanying GitLab repository, see section VIII.

IV. RESULTS AND DISCUSSION

In the following chapter, results are presented as graphs of different measures of congestion over the course of one full simulated day for different values of the probability of closure, p. Each simulation was run with a time step of $\Delta t = 0.1$s and for a total simulated time of 48 hours. The first 24 hours were used as a “warm up” phase and are not included in the results below.

For reference, the number of vehicles on the network at the same time ranged from 500 to 11,000, and the number of routes completed in a full day was around 200,000 to 250,000.

In the first figure, Fig. 2, we see the average vehicle speed over the course of a day. As expected, there is a large correlation between increased probability of road closure and decreased average speed. And during rush hours, the effect is even more severe. Note however that this is a global average of all vehicles on the network, meaning that there is a large spread in speed among the vehicle fleet. Note also that even with zero roads closed, we have an unexpectedly low average speed overall, considering that the speed limit everywhere is 50 km/h. This indicates that even though vehicles never have to reroute, we still have slower moving traffic in heavily trafficked areas (for example, the bridges and tunnels over and under the river flowing through the city).

Another metric of congestion we could analyze is the fraction of vehicles that are in a standstill (in this case meaning a speed lower than 0.1m/s). This metric is plotted in Fig. 3 over the course of a day. Here we see that for no roads closed, there is almost no standstill traffic, except during rush hours, while for higher fractions of road closures we see up to 3% of the vehicles standing still.

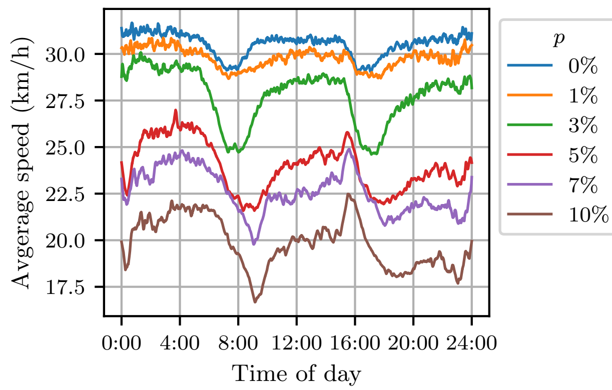

FIG. 2: **Average vehicle speed over a day**. The figure shows the average (instantaneous) speed of all current vehicles on the network as a function of time of day and for different probability of road closure p. Note that a filter has been used on the data to smooth out the graph.

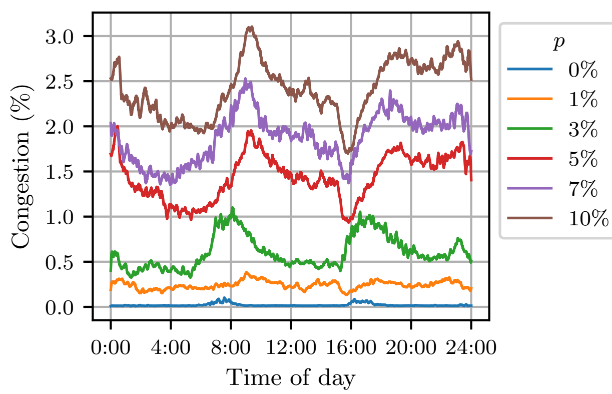

FIG. 3: **Congestion over a day**. The figure shows a metric of congestion (fraction of vehicles that are in stand-still) as a function of time of day and for different probability of road closure p. Note that a filter has been used on the data to smooth out the graph.

Lastly, we will consider the vehicle routes and how their expected travel time differs from the actual time of the route. In Fig. 4, we see the average delay that vehicles experience over the course of a day. As expected, considering the other results, we see a clear increase in delay during rush hours. However, we have one outlier at $p = 7\%$. We would have guessed that the delay would be somewhere between 5% and 10%, but that was not the case. One hypothesis regarding why this happened is that average delay is more dependent on which roads are closed, and not necessarily on how many. Some roads in a road network are more crucial than others, such as main roads connecting larger areas. If such roads happen to be closed, then vehicles will always have to take a detour to reach their destination, without necessarily decreasing their average speed or causing congestion.

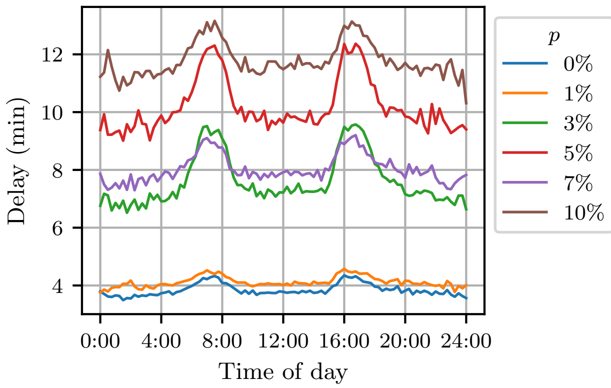

FIG. 4: **Average delay over a day**. The figure shows the average delay (meaning the difference between expected travel time and actual travel time) as a function of time of day and for different probability of road closure p. Note that expected travel time is calculated given no traffic.

V. CONCLUSIONS AND OUTLOOK

The question we were seeking to answer by trying to simulate Gothenburg traffic was: *"How do road closures affect congestion levels in a simplified model of the road network of Gothenburg?"*. The simple answer to that question is: *"When you close more roads, traffic flow will be worse"*, because that is what the graphs in section IV tell us. However, that simple answer reflects only the simplicity of the model. The fact that we only analyzed global congestion metrics, that we closed roads randomly without any consideration of their vitality, and that all roads, highway or small one-way street, were treated as equal, severely restricts how much we can take away from the results.

That said, the agent-based simulation worked, and the fact that the simulation could handle upwards of 11,000 vehicles at a time with (more or less) realistic behavior in a road network of Gothenburg-scale is a success in itself. It is fascinating to see that, even given random spawn points and destinations, we still recognize some of the most trafficked areas in Gothenburg in the simulation as well, such as Korsvägen, Ullevi, Hisingsbron, and Linnéplatsen, to name a few.

Had this project been worked on further, many improvements would have been made. The first step would be to improve the road network with more data, for example speed limits, number of lanes, and street names (to be able to select precise roads to close, for example). Such data could probably be found in the same source that was used in this project, but it was not prioritized at the time. An alternative would be to use OpenStreetMap (OSM) data. That approach was tried in the beginning of the project, but to no success.

Another thing that would be interesting to consider is how the simulated results compare to real world measurements of traffic flow. That was, however, again not prioritized due to limitations in time and resources.

VI. CONTRIBUTIONS

The author (S.E.) developed all code used for the project and produced all the text and figures in the final report.

VII. CONFLICT OF INTEREST

No relevant conflicts of interest were identified before or during the course of the project.

VIII. DATA AND CODE AVAILABILITY

The road network data used in this project is public data and can be downloaded by creating an account at `lastkajen.trafikverket.se` and making a custom order using their tools.

The complete source code for running the simulation and producing the figures in this report, as well as the road network data used in the project, is available on Chalmers Gitlab: `https://git.chalmers.se/simoe/ffr120-project-traffic-simulation-of-gothenburg/`.

[1] J. Nguyen, S. T. Powers, N. Urquhart, T. Farrenkopf, and M. Guckert, An overview of agent-based traffic simulators, Transportation Research Interdisciplinary Perspectives **12**, 10.1016/j.trip.2021.100486 (2021), cited by: 85. Used for: Early inspiration on different types of traffic simulation models and existing frameworks. None of the existing models were ever used during the project.

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